

1. Network Fundamentals

1.1 OSI Model

Each layer serves the layer above it.

7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical

1.2 TCP/IP Model

Application, Transport, Internet, Network Access

Encapsulation adds headers at each layer.

2. Addressing

IPv4: 32-bit, dotted decimal

Subnetting: borrow host bits

/30 gives 2 usable hosts